

Equipment Datasheet — 500-P-505

Sludge Recirculation Pump | ETP Unit

Document ID	DS-500-P-505
Equipment Tag	500-P-505
Equipment Name	Sludge Recirculation Pump
Process Unit	Effluent Treatment Plant (500)
Manufacturer	Cummins India
Install Date	2020-04-22
Design Pressure	115.4 barg
Design Temperature	156.2 degC
Material of Construction	Carbon Steel
Criticality Rating	Medium
Governing SOP	SOP-500-015
Governing Safety Procedure	SP-002

1. Equipment Description

500-P-505 is a Sludge Recirculation Pump installed in the Effluent Treatment Plant (ETP) at PRPC. It was supplied by Cummins India and commissioned on 2020-04-22. The equipment is classified as Medium criticality within the unit's reliability programme.

2. Design Basis

Design pressure is rated at 115.4 barg with a design temperature of 156.2 degC. Primary material of construction is Carbon Steel, selected for compatibility with process fluid composition and corrosion allowance requirements specific to ETP service.

3. Operating & Safety References

Operation of 500-P-505 is governed by procedure SOP-500-015 (Sludge Dewatering and Disposal). Isolation and personnel safety during maintenance follow SP-002 (Hot Work Permit and Fire Watch).

4. Maintenance Philosophy

500-P-505 is included in the ETP rotating/static equipment reliability programme with scheduled preventive maintenance and risk-based inspection (RBI) intervals determined by criticality rating (Medium).